SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SC

COMPUTER SCIENCE AND ENGINEERING

CSA47 – DEEP LEARNING

ASSIGNMENT – 1

CO1 – MATHEMATICAL FOUNDATIONS AND MACHINE LEARNING PRINCIPLES

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Introduction

A healthcare risk prediction system uses patient information such as age, blood pressure, glucose level, cholesterol and family history to estimate disease risk. Reliable learning systems require suitable mathematical representations, probability, numerical computation and careful control of model complexity. This assignment explains these foundations and shows how supervised and unsupervised learning can be used responsibly in a hospital setting.

TASK 1 – Patient Data Representation and Normalization (5 Marks)

a) Using raw features without centering or normalization can cause features with large numerical ranges to dominate calculations. For example, cholesterol values may be around 180–260, while age may be around 25–75. In distance-based methods, the cholesterol difference can contribute more to distance simply because of its scale. In gradient-based methods, differently scaled inputs can produce poorly conditioned optimization and slow or unstable convergence.

b) Consider the following small example containing four measurements for three patients.

Patient	Age	Blood pressure	Glucose	Cholesterol
P1	25	120	90	180
P2	50	130	110	240
P3	75	160	150	210

$$\text{Feature matrix } X = [[25, 120, 90, 180], [50, 130, 110, 240], [75, 160, 150, 210]]$$

A suitable method is Min–Max normalization, which scales each feature to the interval [0, 1]:

$$x_norm = (x - x_min) / (x_max - x_min)$$

The minimum and maximum are calculated separately for each column:

Patient	Normalized age	Normalized BP	Normalized glucose	Normalized cholesterol
P1	0.00	0.00	0.00	0.00
P2	0.50	0.25	0.33	1.00
P3	1.00	1.00	1.00	0.50

Verification example: for P2 glucose, x = 110, minimum = 90 and maximum = 150. Therefore, (110 − 90) / (150 − 90) = 20/60 = 0.333. All normalized values lie between 0 and 1.

c) Normalization is important for distance-based methods such as K-nearest neighbours and K-means because it prevents large-scale features from dominating distance. It is also useful for gradient-based methods such as neural networks and logistic regression because similarly scaled inputs generally make optimization more efficient. Normalization does not remove the need to check outliers or choose an appropriate scaling method.

TASK 2 – Vector Representation and Matrix Operations (2 Marks)

a) Represent one patient using a feature vector and assign a weight to each feature. For illustration, use normalized values in the order age, glucose, blood pressure and cholesterol.

$$\mathbf{x} = [0.50, 0.60, 0.40, 0.70]^T \qquad \mathbf{w} = [0.20, 0.40, 0.20, 0.20]^T$$

$$\text{Weighted score: } S = \mathbf{w}^T\mathbf{x} = 0.20(\text{age}) + 0.40(\text{glucose}) + 0.20(\text{BP}) + 0.20(\text{cholesterol})$$

b) Substituting the values:

$$S = (0.20 \times 0.50) + (0.40 \times 0.60) + (0.20 \times 0.40) + (0.20 \times 0.70)$$

$$S = 0.10 + 0.24 + 0.08 + 0.14 = 0.56$$

The weighted score is 0.56, or 56% of the chosen score range. Glucose contributes the largest weighted component because it has the highest assigned weight. This is an illustrative mathematical score, not a medically validated probability or diagnosis.

TASK 3 – Probability and Conditional Probability (6 Marks)

Define D = patient has the disease and E = patient has elevated glucose.

Group	Elevated glucose	Not elevated	Total
Disease (D)	160	40	200
No disease ($\bar{D}$)	120	680	800
Total	280	720	1000

a) The total number of patients with elevated glucose is 160 + 120 = 280.

$$P(E) = 280 / 1000 = 0.28 = 28\%$$

b) The probability of disease given elevated glucose is:

$$P(D \mid E) = P(D \cap E) / P(E) = 160 / 280 = 0.5714 \approx 57.14\%$$

c) The probability of elevated glucose given disease is:

$$P(E \mid D) = 160 / 200 = 0.80 = 80\%$$

d) Conditional probability helps a hospital estimate how strongly a finding is associated with disease. In this example, 57.14% of patients with elevated glucose have the disease in the sample. Such patients may be prioritized for further assessment according to clinical protocols. However, elevated glucose alone cannot confirm disease; other symptoms, tests and clinical judgement are required.

TASK 4 – Bayesian Statistics for Disease Classification (4 Marks)

Given: prevalence P(D) = 0.10; sensitivity P(+ | D) = 0.90; false-positive rate P(+ | $\bar{D}$) = 0.05; and P($\bar{D}$) = 0.90.

a) Using Bayes' theorem:

$$P(D \mid +) = [P(+ \mid D) \times P(D)] / P(+)$$

$$P(+) = P(+ \mid D)P(D) + P(+ \mid \bar{D})P(\bar{D})$$

$$P(+) = (0.90 \times 0.10) + (0.05 \times 0.90) = 0.09 + 0.045 = 0.135$$

$$P(D \mid +) = 0.09 / 0.135 = 0.6667 \approx 66.67\%$$

Therefore, under the stated assumptions, a patient with a positive result has an estimated 66.67% probability of having the disease.

b) The prior probability P(D) is the disease probability before considering the test result; here it is 10%. The likelihood P(+ | D) describes how likely a positive result is among people who have the disease; here it is 90%. Bayes' theorem combines the prior and likelihood with the evidence to produce the posterior probability.

c) Bayesian reasoning is useful because medical decisions often involve uncertainty and incomplete evidence. It allows new test results to update an earlier risk estimate. The result also demonstrates why prevalence matters: even a reasonably accurate test can produce false positives, particularly when a disease is uncommon. The calculation supports clinical assessment but does not replace diagnosis.

TASK 5 – Entropy and Information Gain (3 Marks)

There are 100 records: 40 high-risk and 60 low-risk. Thus, p(high) = 0.4 and p(low) = 0.6.

$$\text{Entropy } H(S) = -\Sigma\, p_i \log_2(p_i)$$

$$H(S) = -[0.4 \log_2(0.4) + 0.6 \log_2(0.6)]$$

$$H(S) = -[(0.4 \times -1.3219) + (0.6 \times -0.7370)]$$

$$H(S) \approx 0.971 \text{ bits}$$

b) Entropy measures uncertainty or impurity in the class labels. Entropy is zero when all records belong to one class. For two classes, the maximum is one bit when both classes are equally represented. The value 0.971 bits indicates considerable uncertainty because the two classes are relatively balanced.

c) Information gain measures how much a feature reduces entropy after splitting the records. It is calculated as the parent entropy minus the weighted average entropy of the child groups:

$$\text{Information Gain} = H(\text{parent}) - \Sigma\, [\, (|\text{child}| / |\text{parent}|) \times H(\text{child})\,]$$

A decision tree can test possible glucose-level thresholds and calculate the resulting information gain. A threshold that creates purer high-risk and low-risk groups gives higher information gain and may be selected as a useful split. The feature must still be checked for clinical relevance and data leakage.

TASK 6 – Numerical Computation and Distance Measurement (3 Marks)

Patient A = (0.70, 0.80, 0.60) and Patient B = (0.50, 0.75, 0.90), in the order glucose, blood pressure and cholesterol.

$$\text{Euclidean distance } d(A,B) = \sqrt{[(A_1-B_1)^2 + (A_2-B_2)^2 + (A_3-B_3)^2]}$$

$$d = \sqrt{[(0.70-0.50)^2 + (0.80-0.75)^2 + (0.60-0.90)^2]}$$

$$d = \sqrt{[(0.20)^2 + (0.05)^2 + (-0.30)^2]}$$

$$d = \sqrt{[0.04 + 0.0025 + 0.09]} = \sqrt{0.1325} \approx 0.364$$

The distance is approximately 0.364. A smaller distance indicates that the patients have more similar values across the selected features, while a larger distance indicates greater difference. Since the features are normalized, no one feature dominates only because it uses a larger unit or numerical range.

TASK 7 – Bias–Variance Trade-off (4 Marks)

Model	Training error	Validation error	Interpretation
A	High	High	High bias / underfitting
B	Very low	High	High variance / overfitting
C	Low	Low and similar	Good generalization

a) Model A has high bias because it performs poorly on both training and validation data. Model B has high variance because it fits training data very well but performs poorly on unseen validation data. Model C shows the most desirable behaviour because its errors are low and similar.

b) A model with low complexity may be unable to represent important patterns, causing high bias. Increasing complexity can reduce bias, but excessive complexity can make the model sensitive to noise in the training set, increasing variance. The goal is to balance both so that the model generalizes to new patients.

c) For Model A, try a more expressive model, add relevant features, reduce excessive regularization or improve feature representation. For Model B, use regularization, simplify the model, collect more representative training data, use early stopping, and apply cross-validation. Both models should be evaluated on an untouched test set.

TASK 8 – Overfitting, Underfitting and Model Capacity (4 Marks)

a) The simple model that fails to capture disease patterns is underfitting. The moderately complex model that performs consistently is showing appropriate generalization. The highly complex model with near-perfect training performance but poor unseen-patient performance is overfitting.

b) Model capacity is the ability of a learning model to represent different patterns and relationships. A low-capacity model may only learn simple relationships. A higher-capacity model can represent more complex relationships, but it may also learn random fluctuations or noise.

c) Increasing capacity can initially improve performance because the model can capture useful patterns that the simpler model missed. After a point, additional capacity may fit noise and patient-specific details in the training data. Training error may continue to decrease while validation error increases. Validation-based model selection, regularization, early stopping and more representative data help control this problem.

TASK 9 – Supervised Learning and Model Evaluation (3 Marks)

a) This is supervised learning because historical patient records contain input features and known target labels: diseased or disease-free. The algorithm learns a mapping from patient features to the known outcome and uses it to predict labels for new patients.

b) The training set is used to learn model parameters and patterns. The testing set is kept separate and is used after model development to estimate performance on unseen records. A validation set or cross-validation can be used for tuning without repeatedly using the final test set.

c) Accuracy = (number of correct predictions / total test predictions) × 100.

$$\text{Accuracy} = (92 / 100) \times 100 = 92\%$$

The model correctly classified 92 out of 100 test patients. Accuracy alone may be misleading if disease cases are rare, so sensitivity, specificity, precision, F1-score and a confusion matrix should also be examined.

TASK 10 – Unsupervised Learning and Patient Grouping (4 Marks)

a) Unsupervised learning is suitable because the records contain patient measurements but do not have predefined risk labels. The algorithm searches for structure or groups in the data without being given the correct class for each patient.

b) K-means clustering works as follows:

- Choose the number of clusters K.
- Initialize K cluster centres (centroids).
- Calculate the distance from each patient record to each centroid and assign the record to its nearest centroid.
- Recalculate each centroid as the mean of the records assigned to that cluster.
- Repeat assignment and centroid updates until the assignments or centroids change very little.

$$\text{K-means objective: minimize } \Sigma_k \Sigma_{x \in C_k} ||x - \mu_k||^2$$

c) Distance, often Euclidean distance after suitable scaling, measures how close two patient feature vectors are. Records with smaller distances are more similar across the selected measurements and are more likely to be assigned to the same cluster.

d) Clusters may reveal groups such as younger patients with lower measurements or patients with higher glucose and blood pressure. Clinicians can investigate these groups, review their records and decide whether further testing or research is appropriate. A cluster is not automatically a disease category or diagnosis.

TASK 11 – Comparison of Supervised and Unsupervised Learning (4 Marks)

Aspect	Supervised learning	Unsupervised learning
Input data	Features with known labels	Features without labels
Main objective	Learn to predict a target label or value	Discover hidden structure or groups
Typical output	Predicted class or probability	Clusters, groupings or patterns
Example here	Predict diseased / disease-free	Group patients by similar measurements

a) Dataset A is suitable for supervised learning because it includes disease labels that can act as target outputs. Dataset B is suitable for unsupervised learning because it has features but no predefined labels.

b) Supervised learning uses known answers during training and aims to predict outcomes for new records. Unsupervised learning does not use target labels and aims to discover patterns, similarities or clusters. Their outputs and evaluation methods are therefore different.

c) For predicting disease status in a new patient, supervised classification is the recommended primary approach because the hospital has labelled historical examples. A classifier can estimate a class or disease probability. Unlabelled records can additionally be clustered to discover subgroups and support exploration, but cluster membership alone should not be treated as a confirmed disease prediction.

TASK 12 – Design Challenge: Selecting and Adapting a Learning Strategy (8 Marks)

A. Impact Analysis (2 Marks)

a) Increasing the dataset from 10,000 to 1,000,000 records increases the amount of data that must be stored, cleaned, transformed and processed. The feature matrix has many more rows, so memory use and computation time increase. Training may require efficient data pipelines, vectorized operations, mini-batch learning, parallel processing or suitable hardware. Larger datasets can improve the representation of patient variability, but only if the records are reliable and representative.

b) Normalization is needed because age, blood pressure, glucose and cholesterol use different scales. Missing-value handling is essential because many algorithms cannot directly process missing entries and careless deletion may introduce bias. Missingness should be investigated; suitable approaches include median imputation for numeric features, a missingness indicator, or clinically justified methods. Data variability matters because the system must learn across different ages, backgrounds and health profiles rather than memorizing a narrow sample.

B. Evaluation of the Original Approach (2 Marks)

a) A single supervised classifier is not sufficient for both tasks. It can predict disease status using labelled examples, but it does not directly discover natural patient groups in unlabelled data. Patient-group discovery is a clustering or other unsupervised learning task.

b) A supervised model generally learns from records with known labels. If it ignores unlabelled records, potentially useful structure and patterns in those records may be missed. It may also be less representative if labelled records are limited or biased. Unlabelled data cannot simply be treated as labelled; it requires separate analysis or a carefully validated semi-supervised method.

C. Adaptation or Redesign (2 Marks)

a) Use a combined analytical pipeline:

- Data preparation: validate records, remove duplicates, check units and ranges, encode categorical variables, investigate missingness and impute values using training data only.
- Feature scaling: apply a consistent scaler such as Min–Max scaling or standardization, fitted on the training data and then applied to validation, test and new records.
- Supervised branch: train a classifier such as regularized logistic regression, a decision tree or a random forest using labelled patient records to predict disease risk.
- Unsupervised branch: apply K-means or another clustering method to appropriately prepared patient features, including suitable unlabelled records, to identify groups for further investigation.

- Integration and review: present predicted probabilities and cluster membership as separate outputs. Clinicians review results and decide on further tests; neither output independently establishes a diagnosis.

b) Probability can express estimated disease risk and support threshold-based triage, with thresholds selected according to clinical requirements. Distance measures such as Euclidean distance can support K-means after normalization. Model capacity should be controlled using regularization, depth limits, early stopping or cross-validation to avoid overfitting. Missing values must be handled consistently, and scaling parameters must be learned from training data to prevent data leakage.

D. Evidence-Based Defence (2 Marks)

a) Compare the original supervised-only model with the redesigned supervised-plus-clustering pipeline using the same patient-level training, validation and test partitions. Suitable experiments include:

- Train baseline classifiers and the redesigned classifier on the same labelled training records; tune settings using validation data or cross-validation.
- Evaluate classification using sensitivity (recall), specificity, precision, F1-score, ROC-AUC, PR-AUC, calibration and a confusion matrix. Accuracy may also be reported but should not be used alone.
- Evaluate clustering using silhouette score, within-cluster sum of squares (inertia), cluster stability and clinical review of cluster profiles. Clustering quality should not be judged as disease-prediction accuracy when ground-truth labels are unavailable.
- Measure runtime, peak memory use, missing-value rates and performance across relevant patient subgroups. Check for data leakage and evaluate the final selected model once on an untouched test set.

b) The redesigned approach is preferable because it separates two different objectives: supervised classification predicts disease status, while unsupervised clustering explores patient structure. Normalization makes distance calculations more meaningful; probability supports risk estimation; and capacity control improves generalization. At one million records, scalable preprocessing, mini-batch or incremental methods where appropriate, efficient storage and monitoring are important. The final design uses labelled and unlabelled information without pretending that unlabelled records have known outcomes. It should be deployed only after clinical validation, privacy protection, bias assessment and appropriate human oversight.

Conclusion

The mathematical foundations of machine learning directly influence the reliability of healthcare prediction systems. Feature matrices and normalization provide consistent numerical inputs; vectors and matrix operations combine measurements; probability and Bayes' theorem quantify uncertainty; entropy and information gain help evaluate useful features; and distance measures support similarity analysis. Bias–variance balance, model capacity and proper testing are essential for generalization. For the hospital scenario, supervised classification is suitable for predicting disease risk from labelled records, while unsupervised clustering can discover groups in unlabelled records. A combined, carefully evaluated system can support clinical investigation, but it must not replace professional medical judgement or diagnosis.

Important Note

All numerical values and model scores in this assignment are illustrative educational examples. A real healthcare system requires representative clinical data, validated measurement procedures, privacy and security safeguards, fairness testing, calibration, external validation and approval under applicable clinical and regulatory requirements.